README

H1N1 Vaccine Prediction – Dataset B

1. Project Overview

This project uses supervised machine learning classification to predict whether a respondent received the H1N1 vaccine. The dataset contains respondent characteristics, health-related information, opinions about H1N1 and vaccination, and preventive behaviours. The target variable is **h1n1_vaccine**, where 0 represents not vaccinated and 1 represents vaccinated.

2. Dataset Description

Item	Description
Dataset	Dataset B – H1N1 Vaccine Prediction
Identifier	respondent_id (unique random identifier; excluded from model features)
Predictor variables	30 input features, as described in the dataset specification
Target variable	h1n1_vaccine (0 = No, 1 = Yes)
Vaccine cost	$500 per vaccine (given project assumption)
Hospitalisation cost	$25,000 per hospitalisation (given project assumption)

3. Machine Learning Workflow

1. Import Python libraries for data analysis, visualisation, preprocessing, modelling and evaluation.
2. Upload the dataset file and load it into a pandas DataFrame.
3. Explore the data: inspect dimensions, column names, data types, duplicates and missing values.
4. Examine the target distribution to understand the number and proportion of vaccinated and non-vaccinated respondents.
5. Separate predictors (X) from the target (y), excluding respondent_id.
6. Split the data into training and testing sets, using stratification to preserve the target-class proportions.
7. Impute missing numerical and categorical values; encode categorical variables where required.
8. Train and compare Logistic Regression, Decision Tree, Random Forest and Gradient Boosting classifiers.
9. Evaluate models using accuracy, precision, recall, F1-score and ROC-AUC.
10. Inspect the selected model's classification report and confusion matrix, and examine Random Forest feature importance.
11. Compare the stated vaccine and hospitalisation costs as a simple financial scenario.

4. Models and Evaluation Metrics

Models: Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.

Metric	Meaning
Accuracy	Proportion of all predictions that are correct.

Metric	Meaning
Precision	Among respondents predicted as vaccinated, the proportion actually vaccinated.
Recall	Among respondents who were vaccinated, the proportion correctly identified.
F1-score	Harmonic mean of precision and recall.
ROC-AUC	Measures how well the model distinguishes the two target classes across classification thresholds.

5. Running the Notebook

1. Open the notebook in Google Colab or a compatible Jupyter environment.
2. Run the library-import cell.
3. Run the upload cell and select the dataset file from your computer.
4. Run the remaining cells in order, from data exploration through model evaluation.
5. Check that the uploaded file contains the target column named h1n1_vaccine and the identifier column named respondent_id.

Example upload code

```
from google.colab import files
uploaded = files.upload()
file_name = next(iter(uploaded))
df = pd.read_csv(file_name)
display(df.head())
```

6. Important Data Checks and Limitations

- The dataset specification describes 31 columns in total: respondent_id, 29 other input features, and the target h1n1_vaccine. Confirm the actual CSV headers and column count before modelling; the stated feature count should be checked against the supplied file.
- If h1n1_vaccine is missing from the uploaded file, supervised classification cannot be trained until the correct labelled dataset is obtained. Do not create or guess target labels.
- The respondent_id column is an identifier and should not be used as a predictive feature.
- Feature importance describes how a fitted model uses variables for prediction; it does not prove that a feature causes vaccination.
- The financial section compares the supplied costs only. It does not calculate real savings or hospitalisations prevented because vaccine effectiveness and hospitalisation probabilities are not provided.
- A vaccine costing $500 is one-fiftieth of a $25,000 hospitalisation cost. This ratio alone does not establish that vaccinating every respondent is economically beneficial.

7. Expected Outputs

The notebook produces dataset summaries, missing-value analysis, a target-distribution chart, a comparison table and chart for the four classifiers, a classification report, a confusion matrix, a ranked Random Forest feature-importance table and chart, and a basic financial cost comparison. Actual metric values depend on the uploaded dataset and the model training results.

Project documentation prepared for the Dataset B H1N1 Vaccine Prediction notebook.